“Computational Fluid Dynamics I” report No.2

July 2, 2019

Submit this report by 16:00 August 2 (Friday) to the department office.

(1) Suppose that air in a shock tube is divided into left and right sides by a membrane. What flow is generated in the tube after the membrane is broken? Obtain numerical solution as an inviscid flow using one-dimensional Euler equations $\frac{\partial Q}{\partial t}+\frac{\partial E}{\partial x}=0$ and discuss the phenomenon. The initial state in the tube is

Left: P = 1.0, ρ = 1.0, U = 0

Right: P = 0.1, ρ = 0.125, U = 0

(Sod problem). Use MacCormack explicit scheme. Note that you cannot obtain any solution with the MacCormack explicit scheme if explicit numerical viscosity term is not added.

(Hint) Numerical viscosity term, which is often used, is the below, where ε is 2.0.

$$\varepsilon\Delta t\left|\frac{p_{j+1}-2p+p_{j-1}}{p_{j+1}+2p+p_{j-1}}\right|\left(Q_{j+1}-2Q_j+Q_{j-1}\right)$$

(2) Make a program to solve the SOD problem using first-order TVD scheme or similar scheme such as Roe scheme, FVS scheme, Osher scheme, and AUSM scheme. Then, discuss difference from the result obtained in (1). Include the program in the report and describe the points you take care and devise.

(3) I appreciate your feedback to improve the lecture in the next year.